

Module 4: Supervised Learning
Submodule: 4.6 Gaussian Discriminant Analysis
Lecture: Gaussian Discriminant Analysis (GDA)

XCS229 Machine Learning

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I. Introduction

In this lecture, we discuss two instantiations of the general concept:

- **Continuous features** x (GDA)
- **Discrete features** x (spam filtering)

We will primarily focus on the continuous case, which is referred to as **Gaussian Discriminant Analysis** (GDA). In the next lecture, we will cover an application to spam filtering.

II. Gaussian Discriminant Analysis (GDA)

Modeling Assumption

For GDA, we make the following modeling assumption:

$p(x | y)$ follows a Gaussian distribution.

This implies that x given y follows a Gaussian distribution with some mean and covariance. Specifically, we will work with a high-dimensional vector $x \in \mathbb{R}^d$ (for convenience we drop the $x_0 = 1$ convention for the bias term), where the Gaussian distribution can be represented as:

$$x \mid y \sim \mathcal{N}(\mu, \Sigma)$$

where μ is the mean vector and Σ is the covariance matrix.

Multivariate Gaussian Distribution Review

The multivariate normal distribution in d -dimensions, also called the **multivariate Gaussian distribution** $z \sim \mathcal{N}(\mu, \Sigma)$ where $z \in \mathbb{R}^d$ is parameterized by:

- Mean vector: $\mu \in \mathbb{R}^d$.

$$\mathbb{E}[z] = \mu$$

- Covariance matrix: $\Sigma \in \mathbb{R}^{d \times d}$,

$$\text{Cov}(z) \triangleq \mathbb{E}[(z - \mathbb{E}[z])(z - \mathbb{E}[z])^T] = \Sigma$$

The entries in the covariance matrix Σ_{ij} capture the correlation between the i^{th} and j^{th} dimensions. For example:

$$\Sigma_{ij} = \text{Cov}(z_i, z_j) = \mathbb{E}[(z - \mathbb{E}[z])(z - \mathbb{E}[z])^T]$$

If z_1 and z_2 are correlated, their density contours will be ellipses rather than circular shapes, reflecting the direction of the correlation.

The probability density function of a multivariate Gaussian distribution is given by:

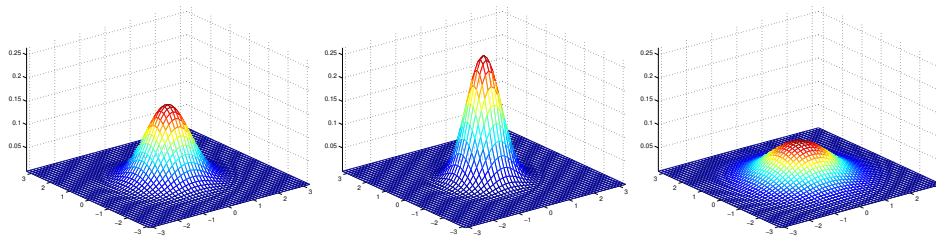
$$p(z) = \frac{1}{(2\pi)^{d/2} \sqrt{|\Sigma|}} e^{-\frac{1}{2}(z-\mu)^T \Sigma^{-1} (z-\mu)}$$

where $|\Sigma|$ is the determinant of the matrix Σ .

The probability density of the Gaussian distribution exhibits certain characteristics:

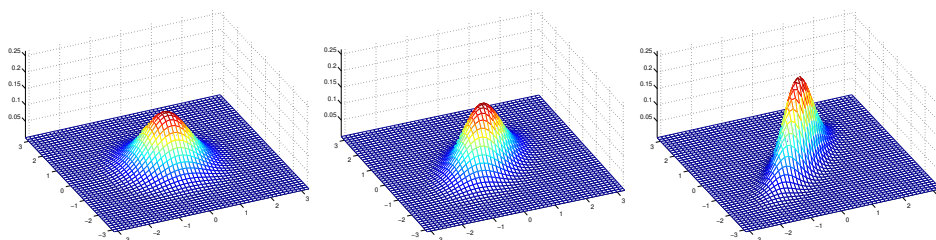
- If Σ is the identity matrix, there are no correlations between dimensions.
- Changes in Σ affect both the size and orientation of the probability density function.

The shapes of these distributions can be visualized in 2D, and the effectiveness of the covariance matrix impacts the density function's characteristics. Here are some examples of what the density of a Gaussian distribution looks like:



- The left-most figure: shows a Gaussian with mean zero (that is, the 2x1 zero-vector) and covariance matrix $\Sigma = I$ (the 2x2 identity matrix). A Gaussian with zero mean and identity covariance is also called the **standard normal distribution**.
- The middle figure: shows the density of a Gaussian with zero mean and $\Sigma = 0.6I$.
- The right-most figure: shows one with $\Sigma = 2I$. We see that as Σ becomes larger, the Gaussian becomes more spread-out, and as it becomes smaller, the distribution becomes more compressed.

Some more examples.



The figures above show Gaussians with mean 0 and covariance matrices respectively:

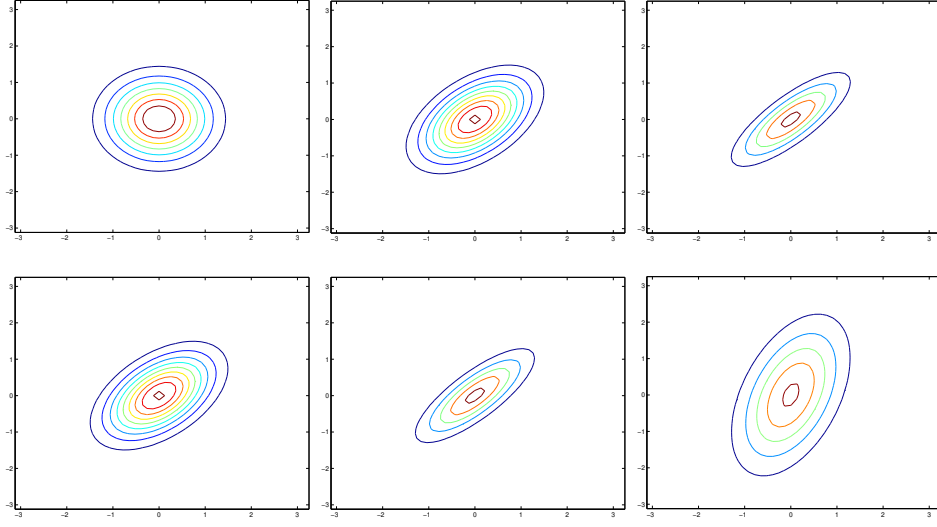
$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad \Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}; \quad \Sigma = \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}.$$

The leftmost figure shows the familiar standard normal distribution, and we see that as we increase the off-diagonal entry in Σ , the density becomes more compressed towards the 45° line (given by $x_1 = x_2$). We can see this more clearly when we look at the contours of the same three densities:

Here's one last set of examples generated by varying Σ :

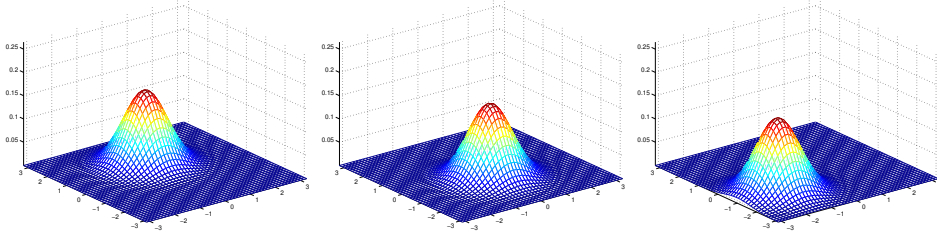
The plots above used, respectively,

$$\Sigma = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}; \quad \Sigma = \begin{bmatrix} 1 & -0.8 \\ -0.8 & 1 \end{bmatrix}; \quad \Sigma = \begin{bmatrix} 3 & 0.8 \\ 0.8 & 1 \end{bmatrix}.$$



From the leftmost and middle figures, we see that by decreasing the off-diagonal elements of the covariance matrix, the density now becomes “compressed” again, but in the opposite direction. Lastly, as we vary the parameters, more generally the contours will form ellipses (the rightmost figure showing an example).

As our last set of examples, fixing $\Sigma = I$, by varying μ , we can also move the mean of the density around.



The figures above were generated using $\Sigma = I$, and respectively

$$\mu = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; \quad \mu = \begin{bmatrix} -0.5 \\ 0 \end{bmatrix}; \quad \mu = \begin{bmatrix} -1 \\ -1.5 \end{bmatrix}.$$

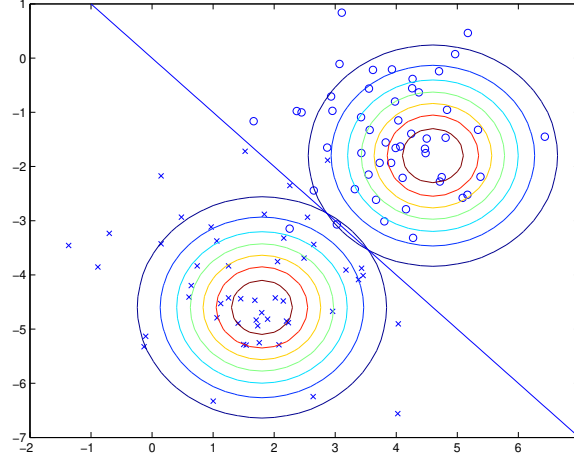
III. Model of GDA

In GDA, we assume discrete labels y . For the classification task, we focus on two labels: 0 and 1.

$$y \in \{0, 1\}$$

Let us consider a case of cancer classification where x_1 and x_2 are measurements. We want to classify cases as benign (0) or malignant (1).

The figure below shows the training set along with the contours of the two Gaussian distributions that have been fitted to the data in each of the two classes.



The two Gaussians have the same shape and orientation since they share a common covariance matrix, Σ . However, the Gaussians have different means, μ_0 and μ_1 , for each class.

Additionally, the figure displays a straight line representing the decision boundary at which $p(y = 1 \mid x) = 0.5$. On one side of the boundary, we predict $y = 1$ as the most likely outcome. On the other side, we predict $y = 0$ as the most likely outcome. This decision boundary is crucial for classification tasks, where the goal is to assign class labels based on the input features.

Parameterizing x given y

We define:

$$\begin{aligned} x \mid (y = 0) &\sim \mathcal{N}(\mu_0, \Sigma) \\ x \mid (y = 1) &\sim \mathcal{N}(\mu_1, \Sigma) \end{aligned}$$

Here μ_0 and μ_1 are the mean vectors for each class $\mu_0, \mu_1 \in \mathbb{R}^d$, while $\Sigma \in \mathbb{R}^{d \times d}$ is the common covariance matrix across both classes.

The prior distribution over the classes $p(y)$ is modeled using parameters:

$$\begin{aligned} \phi &= p(y = 1) \\ 1 - \phi &= p(y = 0) \end{aligned}$$

Thus, y can be treated as a **Bernoulli distribution** with parameter ϕ .

$$y \sim \text{Bernoulli}(\phi)$$

In summary, the parameters we need to learn are:

$$\mu_0, \mu_1, \Sigma \text{ and } \phi$$

Our goal is to learn these parameters so that we can compute $p(y | x)$ using the relationships established.

IV. Summary

In this lecture, we covered the core concepts of **Gaussian Discriminant Analysis (GDA)** with a focus on modeling continuous features using Gaussian distributions. Below are the key points:

- **Modeling Assumptions:** In GDA, we assume that the data for each class follows a Gaussian distribution. Specifically, the features x conditioned on the class y are modeled as multivariate Gaussian distributions with means μ_0 and μ_1 , and a shared covariance matrix Σ .
- **Multivariate Gaussian Distribution:** The Gaussian distribution in d dimensions is parameterized by a mean vector $\mu \in \mathbb{R}^d$ and a covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$. The covariance matrix captures the relationship between the dimensions of the data and affects the shape of the distribution (e.g., how spread out or compressed the contours are).
- **Covariance Matrix Impact:** By adjusting the covariance matrix Σ , we can modify the shape of the Gaussian probability density. A larger value of Σ leads to more spread-out contours, while a smaller Σ results in a more compressed distribution. The off-diagonal elements of Σ influence the correlation between dimensions which is reflected in the orientation of the contours.
- **Decision Boundary:** The decision boundary in GDA is defined as the point where the posterior probability $p(y = 1 | x) = 0.5$. On one side of the boundary, we predict class $y = 1$ and on the other side, we predict $y = 0$. The boundary is determined based on the Gaussian distributions of the two classes and the prior probability of the classes.
- **Learning Parameters:** The goal in GDA is to learn the parameters of the Gaussian distributions, including the mean vectors μ_0 and μ_1 , the covariance matrix Σ , and the prior probability ϕ of class $y = 1$. Once these parameters are learned, we can use Bayes' rule to compute the posterior probability $p(y | x)$ and classify new data points.

- **Gaussian Distributions in Classification:** GDA is particularly useful in classification tasks, such as the cancer classification example, where we aim to distinguish between two classes (e.g., benign vs. malignant) based on input features.

Overall, GDA is a powerful method that allows us to model the distribution of features for each class and use this information to classify new data based on learned parameters.